

Lecture 9

Conjugate priors: Part 1

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(Edited from the lecture notes by Brian J. Reich, NC State)



Conjugate priors

- ▶ A prior is called **conjugate** if the posterior belongs to the same parametric family as that of the prior.
- ▶ We have seen that in case of a binomial likelihood and a beta prior, the posterior is also a beta.
- ▶ This requires a mathematical pairing of the likelihood and prior.
- ▶ There is a long list of conjugate priors
https://en.wikipedia.org/wiki/Conjugate_prior
- ▶ The advantage of a conjugate prior is that the posterior has a closed form expression.

Why conjugate priors?

- ▶ Let us see an example of a non-conjugate prior.
- ▶ Suppose, $Y \sim \text{Poisson}(\lambda)$ with $\lambda \in (0, 1)$, and $\lambda \sim \text{Beta}(a, b)$.
- ▶ The posterior is

$$f(\lambda|Y) \propto \left\{ \exp(-\lambda) \lambda^Y \right\} \left\{ \lambda^{a-1} (1 - \lambda)^{b-1} \right\} \mathbb{1}(\lambda \in [0, 1])$$

- ▶ This is not a beta PDF, so the prior is not conjugate.
- ▶ In fact, this is not a member of any known family of distributions.
- ▶ For some model parameters, there is no known conjugate prior.

Why conjugate priors?

- ▶ The final form of the posterior is

$$f(\lambda|Y) = \frac{\{\exp(-\lambda)\lambda^Y\} \{\lambda^{a-1}(1-\lambda)^{b-1}\}}{\int_0^1 \{\exp(-\lambda_0)\lambda_0^Y\} \{\lambda_0^{a-1}(1-\lambda)^{b-1}\} d\lambda_0} \mathbb{1}(\lambda_0 \in [0, 1])$$

- ▶ To calculate the posterior mean, you need to evaluate

$$E(\lambda|Y) = \frac{\int_0^1 \lambda_0 \{\exp(-\lambda)\lambda^Y\} \{\lambda^{a-1}(1-\lambda)^{b-1}\} d\lambda_0}{\int_0^1 \{\exp(-\lambda_0)\lambda_0^Y\} \{\lambda_0^{a-1}(1-\lambda)^{b-1}\} d\lambda_0}.$$

- ▶ Impossible to evaluate without a software.

Example of a conjugate prior

Estimating a proportion using the beta/binomial model

- ▶ A fundamental task in statistics is to estimate a proportion using a series of trials:
 - ▶ What is the success probability of a new cancer treatment?
 - ▶ What proportion of voters support my candidate?
 - ▶ What proportion of the population has a rare gene?
- ▶ Let $\theta \in [0, 1]$ be the proportion we are trying to estimate (e.g., the success probability).
- ▶ We conduct n independent trials, each with success probability θ , and observe $Y \in \{0, \dots, n\}$ successes.
- ▶ We would like obtain the posterior of θ , a 95% interval, and a test that θ equals some predetermined value θ_0 .

Frequentist analysis

- ▶ The maximum likelihood estimate is the sample proportion

$$\hat{\theta} = Y/n.$$

- ▶ For large Y and $n - Y$, the sampling distribution of $\hat{\theta}$ is approximately

$$\hat{\theta} \sim \text{Normal} \left(\theta, \frac{\theta(1 - \theta)}{n} \right).$$

- ▶ The standard error (standard deviation of the sampling distribution) is approximated as

$$\text{SE}(\hat{\theta}) \approx \sqrt{\frac{\hat{\theta}(1 - \hat{\theta})}{n}}.$$

- ▶ A 95% CI is then

$$\hat{\theta} \pm 2\text{SE}(\hat{\theta}).$$

Bayesian analysis - Likelihood

- ▶ Since Y is the number of successes in n independent trials, each with success probability θ , its distribution is

$$Y|\theta \sim \text{Binomial}(n, \theta).$$

- ▶ PMF: $P(Y = y|\theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$.
- ▶ Mean: $E(Y|\theta) = n\theta$.
- ▶ Variance: $V(Y|\theta) = n\theta(1 - \theta)$.

Bayesian analysis - Prior

- ▶ The parameter θ can take all possible values between 0 and 1. Therefore, a natural prior is

$$\theta \sim \text{Beta}(a, b).$$

- ▶ PDF: $f(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1 - \theta)^{b-1}.$

- ▶ Mean: $E(\theta) = \frac{a}{a+b}.$

- ▶ Variance: $V(\theta) = \frac{ab}{(a+b)^2(a+b+1)}.$

Derivation of the posterior

- ▶ The posterior is

$$\begin{aligned} p(\theta|Y) &\propto f(Y|\theta)\pi(\theta) \\ &\propto \theta^Y(1-\theta)^{n-Y} \times \theta^{a-1}(1-\theta)^{b-1} \mathbb{1}(\theta \in [0, 1]) \\ &\propto \theta^{Y+a-1}(1-\theta)^{n-Y+b-1} \mathbb{1}(\theta \in [0, 1]). \end{aligned}$$

- ▶ By matching the form of the kernel with that of a beta distribution, the posterior is

$$\theta|Y \sim \text{Beta}(a+Y, b+n-Y).$$

- ▶ Our prior choice for θ was beta and the posterior we obtain is again beta.
- ▶ Thus, $\text{Beta}(a, b)$ is a conjugate prior for θ .

Shrinkage

- ▶ The posterior mean is

$$\hat{\theta}_B = E(\theta|Y) = \frac{y + a}{n + a + b}.$$

- ▶ The posterior mean is between the sample proportion Y/n and the prior mean $a/(a + b)$:

$$\hat{\theta}_B = w \frac{Y}{n} + (1 - w) \frac{a}{a + b},$$

where the weight on the sample proportion is $w = \frac{n}{n + a + b}$.

- ▶ $\hat{\theta}_B$ close to Y/n when a and b are negligible in comparison of n .
- ▶ $\hat{\theta}_B$ shrinks towards the prior mean $a/(a + b)$ when n is negligible in comparison of $a + b$.

Selecting the prior

- ▶ The posterior is $\theta|Y \sim \text{Beta}(a + Y, b + n - Y)$.
- ▶ Therefore, a and b can be interpreted as the “prior number of success and failures”.
- ▶ If we have no information about θ before collecting data, we should select a and b to be tiny in comparison of n .
- ▶ If historical data/expert opinion indicates that θ is likely between 0.6 and 0.8, we should select a and b in such a way that $\text{Prob}(0.6 < \theta < 0.8)$ is high.

Related problem

- ▶ Suppose, you are playing a computer game where you have n (known) lives.
- ▶ In each turn, either you win a point or you lose a life (computer wins).
- ▶ You keep on playing until you lose all n lives and suppose your score is Y .
- ▶ The success probability of independent stages of the game is θ .
- ▶ Y is the number of successes before we observe n failures.
- ▶ Then $Y|\theta \sim \text{Negative-Binomial}(n, \theta)$ and

$$\text{Prob}(Y = y|\theta) = \binom{y+n-1}{y} \theta^y (1-\theta)^n.$$

- ▶ Assume the prior $\theta \sim \text{Beta}(a, b)$ and find the posterior.
- ▶ **R code:** <https://stat.ethz.ch/R-manual/R-devel/library/stats/html/NegBinomial.html>

Derivation of the posterior

- ▶ The posterior is

$$\begin{aligned} p(\theta|Y) &\propto f(Y|\theta)\pi(\theta) \\ &\propto \theta^Y(1-\theta)^n \times \theta^{a-1}(1-\theta)^{b-1} \mathbb{1}(\theta \in [0, 1]) \\ &\propto \theta^{Y+a-1}(1-\theta)^{n+b-1} \mathbb{1}(\theta \in [0, 1]). \end{aligned}$$

- ▶ By matching the form of the kernel with that of a beta distribution, the posterior is

$$\theta|Y \sim \text{Beta}(a+Y, b+n).$$

- ▶ Our prior choice for θ was beta and the posterior we obtain is again beta.
- ▶ Thus, $\text{Beta}(a, b)$ is a conjugate prior for θ .

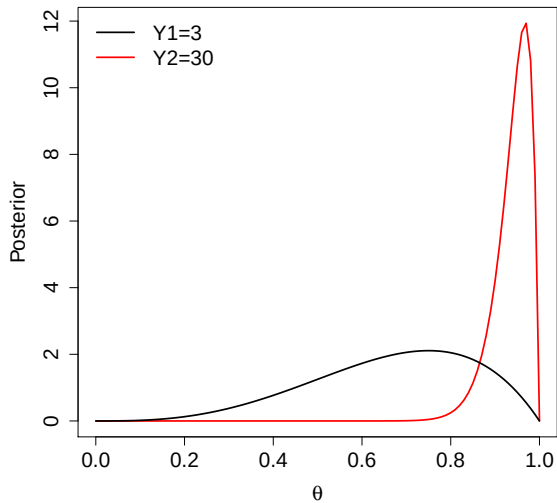
Smoking example

- ▶ Two smokers have just quit.
- ▶ Say subject i has probability θ_i of abstaining each day.
- ▶ The number of days until relapse for two patients is 3 and 30 days.
- ▶ Can we conclude the patients have different probabilities of relapse?
- ▶ What is probability that their next attempts will exceed 30 days?

Smoking example

- ▶ The likelihood is $Y_i \sim \text{Negative-Binomial}(1, \theta_i)$.
- ▶ Assume uniform priors $\theta_i \sim \text{Beta}(1, 1)$.
- ▶ The posteriors are $\theta_i | Y_i \sim \text{Beta}(Y_i + 1, 2)$.
- ▶ The posterior are plotted on the next slide.
- ▶ The following slide uses Monte Carlo sampling to address the two motivating questions.

Smoking example



Smoking example

```
> S <- 1000000
> theta1 <- rbeta(S, 3+1, 2); theta2 <- rbeta(S, 30+1, 2)
> mean(theta2 > theta1)
[1] 0.957222
>
> samp1 <- rbinom(S,1,prob=1-theta1)
> samp2 <- rbinom(S,1,prob=1-theta2)
> quantile(samp1,c(0.05,0.5,0.95))
5% 50% 95%
0    1   15
> quantile(samp2,c(0.05,0.5,0.95))
5% 50% 95%
0   13 109
> mean(samp1>30); mean(samp2>30)
[1] 0.015781
[1] 0.254129
```

Related problem

- ▶ Virat Kohli captained Team India for 68 tests. The team won 40 matches, lost 17 matches and 11 matches were draw.
- ▶ Now, we have more than two categories.
- ▶ The winning, losing, and draw probabilities of independent matches are θ_1 , θ_2 , and θ_3 , respectively. We must have $\theta_1 + \theta_2 + \theta_3 = 1$. Let $\theta = (\theta_1, \theta_2, \theta_3)'$.
- ▶ $\mathbf{Y} = (Y_1, Y_2, Y_3)'$ is the vector of counts of different categories out of n outcomes.
- ▶ Then $\mathbf{Y}|\theta \sim \text{Multinomial}(n, \theta)$ and for $\mathbf{y} = (y_1, y_2, y_3)'$,

$$\text{Prob}(Y = \mathbf{y}|\theta) = \frac{n!}{y_1!y_2!y_3!} \theta_1^{y_1} \theta_2^{y_2} \theta_3^{y_3}.$$

- ▶ **R code:** <https://stat.ethz.ch/R-manual/R-devel/library/stats/html/Multinom.html>

Bayesian analysis - Prior

- ▶ The parameters θ_1 , θ_2 , and θ_3 can take all possible values between 0 and 1 along with the constraint $\theta_1 + \theta_2 + \theta_3 = 1$. Therefore, a natural prior is

$$\theta \sim \text{Dirichlet}(a_1, a_2, a_3).$$

- ▶ The density function is

$$\pi(\theta) = \frac{\Gamma(a_1 + a_2 + a_3)}{\Gamma(a_1)\Gamma(a_2)\Gamma(a_3)} \theta_1^{a_1-1} \theta_2^{a_2-1} \theta_3^{a_3-1}.$$

- ▶ R code: `https://www.rdocumentation.org/packages/DirichletReg/versions/0.6-3/topics/Dirichlet`
- ▶ Mean: $E(\theta_i) = \frac{a_i}{\sum_i a_i}.$
- ▶ Variance: $V(\theta_i) = \frac{a_i}{\sum_i a_i} \times \left(1 - \frac{a_i}{\sum_i a_i}\right) \times \frac{1}{(1 + \sum_i a_i)}.$

Derivation of the posterior

- ▶ The posterior is

$$\begin{aligned} p(\boldsymbol{\theta}|\mathbf{Y}) &\propto f(\mathbf{Y}|\boldsymbol{\theta})\pi(\boldsymbol{\theta}) \\ &\propto \theta_1^{Y_1}\theta_2^{Y_2}\theta_3^{Y_3} \times \theta_1^{a_1-1}\theta_2^{a_2-1}\theta_3^{a_3-1} \\ &\propto \theta_1^{Y_1+a_1-1}\theta_2^{Y_2+a_2-1}\theta_3^{Y_3+a_3-1}. \end{aligned}$$

- ▶ By matching the form of the kernel with that of a Dirichlet distribution, the posterior is

$$\boldsymbol{\theta}|\mathbf{Y} \sim \text{Dirichlet}(Y_1 + a_1, Y_2 + a_2, Y_3 + a_3).$$

- ▶ Our prior choice for $\boldsymbol{\theta}$ was Dirichlet and the posterior we obtain is again Dirichlet.
- ▶ Thus, $\text{Dirichlet}(a_1, a_2, a_3)$ is a conjugate prior for $\boldsymbol{\theta}$.

Thank you!